

The Bitcoin Harmonic Time Model: Internal Phase Symmetry, Amplitude Power-Laws, and the Institutional ETF Regime Shift

Romero Quant

Independent Quantitative Research

`romero.quant@proton.me`

[@RomeroQuant](#)

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Abstract

This paper formalises the Bitcoin Harmonic Time Model, a two-dimensional quantitative framework that structures Bitcoin market cycles into a discrete dynamic system defined by temporal phase symmetry and compressed amplitude distributions. We demonstrate that the governing clock of the market is not the halving event in isolation, but a stable internal phase architecture where the halving acts as a temporal node, consistently arriving at normalised phase $\bar{\phi}(H) \approx 0.494$ of the expansion phase. Using historical data from three completed cycles, we establish that expansion lasts $1,060 \pm 12$ days and contraction 370 ± 13 days, yielding a mean period of 1,430 days. Furthermore, we derive an empirical power-law demonstrating that cycle multipliers scale non-linearly with prior drawdown depth ($\hat{\mathcal{M}}(d) = 328 \cdot d^{5.1}$), and introduce a structural adjustment framework to account for the regime shift initiated by the launch of spot US Exchange-Traded Funds (ETFs) in 2024. Finally, we translate this unified model into explicit, dated, and falsifiable calendar hypotheses for the current and subsequent market cycles.

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1 Empirical Data and Temporal Methodology

1.1 Identification of Cycle Extrema

The empirical foundation of this model rests on the precise identification of two classes of structural inflection points in Bitcoin’s price history: absolute *cycle troughs* (T_i) and absolute *cycle peaks* (P_i), where the subscript $i \in \{1, 2, 3\}$ denotes the cycle index. These inflection points are defined under a strict methodological protocol designed to minimise look-ahead bias and subjective interpolation.

Definition 1 (Cycle Trough). A *cycle trough* T_i is defined as the date of the global price minimum recorded on daily closing data between two consecutive cycle peaks P_{i-1} and P_i , where no subsequent retracement to or below that price level occurs within the same inter-peak interval.

Definition 2 (Cycle Peak). A *cycle peak* P_i is defined as the date of the global all-time high (ATH) closing price recorded within the corresponding expansion phase, measured from trough T_i and bounded above by the subsequent trough T_{i+1} .

By construction, each observation window captures one complete *expansion phase* $[T_i, P_i]$ and one complete *contraction phase* $[P_i, T_{i+1}]$, forming the elementary cycle unit $\mathcal{C}_i = [T_i, P_i, T_{i+1}]$.

The halving dates H_i are treated as exogenous and protocol-determined variables. They are taken from the public blockchain record without adjustment:

- H_1 : 9 July 2016 (block 420,000; reward reduced from 25 to 12.5 BTC)
- H_2 : 11 May 2020 (block 630,000; reward reduced from 12.5 to 6.25 BTC)
- H_3 : 20 April 2024 (block 840,000; reward reduced from 6.25 to 3.125 BTC)

All duration measures Δ_ϕ are expressed in elapsed calendar days, computed as:

$$\Delta_\phi(A, B) = |\text{date}(B) - \text{date}(A)| \quad \text{in days} \quad (1)$$

where A and B are any two dated events in the sequence $\{T_i, H_i, P_i\}$.

Note on Cycle 3. The peak date for the current cycle ($P_3 = 6$ October 2025) is treated as a *working hypothesis* consistent with the temporal structure derived in this section. Its inclusion in the averaged parameters is flagged accordingly throughout the analysis, and all projections derived from it are explicitly labelled as probabilistic. The model is designed to be falsifiable and self-correcting: if a higher closing price is established after the submission date of this paper, the cycle extremum and all downstream projections must be revised.

1.2 Cycle Phase Durations

Table 1 reports the duration of the expansion and contraction phases for each observed cycle. The *expansion duration* Δ_i^+ is measured from trough T_i to peak P_i ; the *contraction duration* Δ_i^- is measured from peak P_i to the subsequent trough T_{i+1} .

Table 1: Bitcoin Market Cycle Phase Durations (Calendar Days)

Cycle	Expansion Phase Δ_i^+ (days)	Contraction Phase Δ_i^- (days)
Cycle 1	1068	363
Trough T_1 : 14 Jan 2015		
Peak P_1 : 17 Dec 2017		
Trough T_2 : 15 Dec 2018		
Cycle 2	1061	376
Trough T_2 : 15 Dec 2018		
Peak P_2 : 10 Nov 2021		
Trough T_3 : 21 Nov 2022		
Cycle 3 [†]	1050	—
Trough T_3 : 21 Nov 2022		
Peak P_3 : 6 Oct 2025 [†]		
Average $\bar{\Delta}$	1060	370
Range [min, max]	[1050, 1068]	[363, 376]
Inter-cycle deviation $\sigma_{\max}$	18 days	13 days

[†] Cycle 3 peak (P_3) is a working hypothesis. The contraction phase Δ_3^- is therefore excluded from the current average computation and will be incorporated upon confirmation of the cycle extremum.

The arithmetic mean of the expansion phase across three cycles yields $\bar{\Delta}^+ \approx 1,060$ days, with a maximum inter-cycle deviation of $\sigma_{\max}^+ = 18$ days—representing a relative dispersion of approximately 1.7% around the mean. The contraction phase, measured over two complete observations, yields $\bar{\Delta}^- \approx 370$ days with a maximum deviation of $\sigma_{\max}^- = 13$ days, a relative dispersion of 3.5%.

1.3 Internal Phase Symmetry: The Halving as Temporal Anchor

The most structurally significant finding of this empirical analysis emerges when the expansion phase is decomposed at the halving event H_i . Rather than treating the halving as the origin of the cycle—as conventional halving-centric models implicitly do—we position it as an internal marker that divides the expansion phase $[T_i, P_i]$ into two measurable sub-intervals:

$$\alpha_i = \Delta_\phi(T_i, H_i) \quad (\text{pre-halving segment}) \quad (2)$$

$$\beta_i = \Delta_\phi(H_i, P_i) \quad (\text{post-halving segment}) \quad (3)$$

such that, by construction, $\alpha_i + \beta_i = \Delta_i^+$.

[Internal Phase Symmetry] The halving event H_i functions as a *temporal anchor* within the expansion phase: the sub-intervals α_i and β_i are quasi-equal and exhibit inter-cycle stability that is statistically independent of price-level dynamics.

Table 2 reports the measured values of α_i and β_i for each observed cycle.

Table 2: Internal Phase Symmetry — Expansion Phase Decomposed at the Halving

Cycle	Halving Date (H_i)	α_i (days)	β_i (days)	$\alpha_i + \beta_i$ (days)
Cycle 1 (2015–2017)	9 Jul 2016	542	526	1,068
Cycle 2 (2018–2021)	11 May 2020	513	548	1,061
Cycle 3 (2022–2025) [†]	20 Apr 2024	515	534	1,050
Average $\bar{\alpha}$, $\bar{\beta}$	—	523	536	1059
Max. deviation $\sigma_{\max}$	—	19 days	12 days	—

[†] Cycle 3 values are derived under the hypothesis $P_3 = 6$ October 2025.

The data reveal a bilateral stability that is difficult to attribute to stochastic coincidence. The pre-halving segment α_i ranges from 513 to 542 days, yielding a mean of $\bar{\alpha} = 523$ days and a maximum deviation of $\sigma_{\max}^{\alpha} = 19$ days ($\approx 3.6\%$ of the mean). The post-halving segment β_i ranges from 526 to 548 days, yielding a mean of $\bar{\beta} = 536$ days with a maximum deviation of $\sigma_{\max}^{\beta} = 12$ days ($\approx 2.2\%$ of the mean). Taken together, the sum $\bar{\alpha} + \bar{\beta} = 1,059$ days is consistent to within one day of the independently computed expansion mean $\bar{\Delta}^+ = 1,060$ days, providing internal cross-validation of the measurement methodology.

The asymmetry between $\bar{\alpha}$ and $\bar{\beta}$ is also noteworthy: at 13 days, it indicates that the halving does not bisect the expansion phase exactly at its midpoint, but rather occurs *slightly before* it. In other words, the system reaches its peak approximately 13 days *later* in the post-halving segment than what a perfectly symmetric division would predict. This mild asymmetry is stable across all three cycles and may reflect the lagged transmission mechanism through which supply reduction propagates into speculative price discovery.

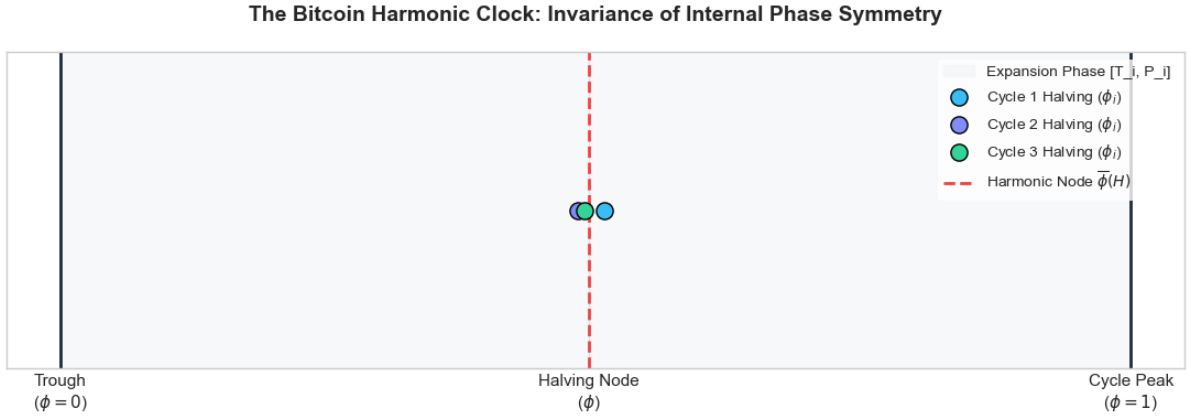


Figure 1: **The Bitcoin Harmonic Clock:** Visualisation of the internal phase architecture across Cycles 1, 2, and 3. The diagram demonstrates the structural invariance of the expansion phase, positioning the halving event not as an origin, but as an internal temporal node fixed at $\bar{\phi}(H) \approx 0.494$.

1.4 Statistical Characterisation of Temporal Stability

The preceding measurements permit a formal statistical characterisation of the temporal regularity of Bitcoin’s market structure. Let us define the *temporal stability index* $\mathcal{S}(\phi)$ of a phase ϕ as:

$$\mathcal{S}(\phi) = 1 - \frac{\sigma_{\max}(\phi)}{\Delta(\phi)} \quad (4)$$

where $\sigma_{\max}(\phi)$ is the maximum observed deviation from the phase mean and $\bar{\Delta}(\phi)$ is the phase mean duration. A value of $\mathcal{S}(\phi) = 1$ would denote perfect invariance; values approaching zero indicate high temporal dispersion.

Applying this definition to the empirically observed phases:

$$\mathcal{S}(\Delta^+) = 1 - \frac{18}{1060} \approx 0.983 \quad (5)$$

$$\mathcal{S}(\Delta^-) = 1 - \frac{13}{370} \approx 0.965 \quad (6)$$

$$\mathcal{S}(\alpha) = 1 - \frac{19}{523} \approx 0.964 \quad (7)$$

$$\mathcal{S}(\beta) = 1 - \frac{12}{536} \approx 0.978 \quad (8)$$

All four phase parameters exhibit temporal stability indices in excess of 0.96. This is a remarkable result for a market that is widely characterised as chaotic and subject to reflexive price dynamics. Most notably, the full expansion phase Δ^+ returns the highest stability index of the four, $\mathcal{S}(\Delta^+) \approx 0.983$, implying that the bullish trajectory from cycle trough to cycle peak is the most temporally invariant structural unit in the system.

Proposition 1 (Differential Temporal Stability). *The expansion phase of Bitcoin market cycles is more temporally stable than the contraction phase: $\mathcal{S}(\Delta^+) > \mathcal{S}(\Delta^-)$. Equivalently, the maximum inter-cycle deviation of the expansion phase (18 days over a mean of 1,060 days) is proportionally smaller than that of the contraction phase (13 days over a mean of 370 days).*

This proposition carries a significant analytical implication. The prevailing market intuition often treats bear markets as more structurally predictable than bull markets—a view shaped by the observation that fear-driven capitulation tends to be temporally compressed and abrupt. The data presented here challenge that intuition at the cycle level: while individual contraction events may feel decisive, their total duration is less invariant across cycles than that of the expansion phases that precede them. Put differently, *the system’s internal clock runs more precisely during expansions than during contractions*.

This asymmetry may reflect a fundamental difference in the dynamics governing the two phases. The expansion phase is characterised by a progressive and relatively orderly accumulation of demand—a process that, by its nature, unfolds over extended timeframes and is anchored by the halving’s structural signal. The contraction phase, by contrast, is governed by the interaction between forced selling, liquidity withdrawal, and sentiment deterioration: forces that are inherently more sensitive to exogenous shocks and therefore more variable in their temporal manifestation.

Critically, the observation that the expansion phase post-ETF approval (Cycle 3) continues to conform to the historical temporal envelope—with a duration of $\Delta_3^+ = 1,050$ days, only 10 days below the prior minimum of 1,061 days—provides preliminary evidence that the temporal structure of the model is robust to the structural regime change introduced by institutional participation. The amplitude parameters, as we discuss in the following sections, do not appear to enjoy the same degree of cross-regime stability.

2 Mathematical Formalisation of Internal Phase Symmetry

2.1 Bitcoin Cycles as a Damped Harmonic System

The empirical regularities documented in Section 2 motivate a formal analogical framework borrowed from the theory of dynamic systems. We propose that the observable structure of Bitcoin market cycles admits a tractable representation as a *discrete damped harmonic oscillator*:

a system characterised by near-constant periodicity, a stable internal phase relationship, and monotonically decreasing amplitude across successive cycles.

In continuous-time mechanics, a damped harmonic oscillator obeys the second-order linear differential equation:

$$\ddot{x}(t) + 2\zeta\omega_0 \dot{x}(t) + \omega_0^2 x(t) = 0 \quad (9)$$

where $x(t)$ denotes the displacement from equilibrium, ω_0 the natural angular frequency of the undamped system, and $\zeta \in (0, 1)$ the damping ratio. For $\zeta \in (0, 1)$ —the underdamped regime—the general solution takes the form:

$$x(t) = A_0 e^{-\zeta\omega_0 t} \cos(\omega_d t + \varphi_0) \quad (10)$$

where $\omega_d = \omega_0 \sqrt{1 - \zeta^2}$ is the damped angular frequency, A_0 is the initial amplitude, and φ_0 is the initial phase offset. The solution describes oscillations of progressively decreasing amplitude enveloped by the exponential decay factor $e^{-\zeta\omega_0 t}$.

We do not claim that equation (9) governs Bitcoin’s price path in any literal mechanical sense. Markets are reflexive, non-linear systems, and price dynamics are not solutions to differential equations with fixed coefficients. The value of the analogy lies in its structural correspondence: (i) a measurable and nearly constant period T ; (ii) a well-defined phase relationship pinned by an internal reference event; and (iii) a monotonically decreasing amplitude envelope. All three properties are empirically supported by the data in Section 2 for the temporal dimension and are developed for the amplitude dimension in Section 4.

2.2 Cycle Period and Angular Frequency

The total cycle period T_{cycle} is defined as the elapsed time between two consecutive troughs:

$$T_{\text{cycle}} = \Delta_i^+ + \Delta_i^- \quad (11)$$

Using the empirically derived averages $\bar{\Delta}^+ \approx 1,060$ days and $\bar{\Delta}^- \approx 370$ days:

$$\bar{T}_{\text{cycle}} \approx 1,060 + 370 = 1,430 \text{ days} \quad (12)$$

The corresponding mean angular frequency $\bar{\omega}$, expressed in radians per day, is:

$$\bar{\omega} = \frac{2\pi}{\bar{T}_{\text{cycle}}} = \frac{2\pi}{1,430} \approx 4.394 \times 10^{-3} \text{ rad} \cdot \text{day}^{-1} \quad (13)$$

2.3 Phase Definition and the Halving as a Temporal Node

Within each cycle $\mathcal{C}_i = [T_i, P_i, T_{i+1}]$, we define the instantaneous cycle phase $\phi(t)$ as the normalised position of any date t within the expansion interval $[T_i, P_i]$:

$$\phi(t) = \frac{t - T_i}{\Delta_i^+}, \quad \phi \in [0, 1] \quad (14)$$

Under this normalisation, $\phi(T_i) = 0$ at the cycle trough and $\phi(P_i) = 1$ at the cycle peak. The phase at the halving event H_i is therefore:

$$\phi(H_i) = \frac{\alpha_i}{\Delta_i^+} = \frac{\alpha_i}{\alpha_i + \beta_i} \quad (15)$$

Substituting the empirical averages $\bar{\alpha} = 523$ days and $\bar{\beta} = 536$ days:

$$\bar{\phi}(H) = \frac{523}{523 + 536} = \frac{523}{1,059} \approx 0.4939 \quad (16)$$

The halving consistently occurs at approximately $\bar{\phi}(H) \approx 0.494$ of the expansion phase—marginally before the mathematical midpoint $\phi = 0.500$. We define the *phase displacement* δ_ϕ as the deviation of the halving from perfect bisection:

$$\delta_\phi = \bar{\phi}(H) - 0.500 = 0.494 - 0.500 = -0.006 \quad (17)$$

This displacement of -0.006 in normalised phase corresponds to approximately $-0.006 \times 1,059 \approx -6$ calendar days: the halving arrives, on average, six days earlier than the geometric midpoint of the expansion. The sign is negative because the system’s peak lies slightly further in time from the halving than the trough does—a structural asymmetry we interpret as the lagged propagation of the supply shock through market sentiment and price discovery mechanisms.

The observed inter-cycle values of $\phi(H_i)$ are:

$$\phi_1(H_1) = \frac{542}{1068} \approx 0.508 \quad (18)$$

$$\phi_2(H_2) = \frac{513}{1061} \approx 0.484 \quad (19)$$

$$\phi_3(H_3) = \frac{515}{1050} \approx 0.490 \quad (20)$$

The range of $\phi(H_i)$ is $[0.484, 0.508]$, a span of 0.024 in normalised phase, or approximately 25 calendar days when projected onto the mean expansion duration. This near-invariance of the halving’s phase position across three independent cycles—each occurring in markedly different macroeconomic and structural environments—constitutes the strongest quantitative argument for treating the halving as a *temporal node*: a fixed point in the phase space of the system, rather than an event whose timing relative to the cycle is coincidental.

Definition 3 (Temporal Node). The halving event H_i is a *temporal node* of cycle $\mathcal{C}_i$ if and only if its normalised phase $\phi(H_i)$ is invariant across cycles up to a maximum deviation of ε_ϕ , where ε_ϕ is small relative to unity. Empirically, $\varepsilon_\phi \approx 0.012$ (one half of the observed range), satisfying the condition $\varepsilon_\phi \ll 1$.

2.4 Implications for Forward Projection

The phase-anchoring property of the halving has a direct and practically consequential implication: knowing the date of the next halving H_{i+1} and an estimate of the trough T_{i+1} , it is possible to construct a probabilistic interval for the next peak P_{i+1} without reference to any price-level information. Specifically:

$$\hat{P}_{i+1} = H_{i+1} + \bar{\beta} \pm \sigma_{\max}^\beta = H_{i+1} + 536 \pm 12 \text{ days} \quad (21)$$

This formulation renders the temporal projection *falsifiable* in a strictly Popperian sense: it generates a dated prediction interval that can be confirmed or refuted by subsequent market data, independently of any assumptions about price magnitude. The projection for the current cycle is developed in full in Section 6.

3 The Law of Amplitude and Structural Regime Adjustments

3.1 Empirical Amplitude Data

The second measurable degree of freedom in the harmonic model is the *amplitude* of each cycle: the magnitude of the price return from trough to peak. We characterise this dimension through

two complementary empirical series. The first is the *cycle multiplier* $\mathcal{M}_i$, defined as the ratio of the peak price to the trough price within cycle i :

$$\mathcal{M}_i = \frac{P_i^\$}{T_i^\$} \quad (22)$$

The second is the *prior drawdown* d_i , defined as the proportional decline from the preceding cycle peak $P_{i-1}^\$$ to the current trough $T_i^\$$:

$$d_i = \frac{P_{i-1}^\$ - T_i^\$}{P_{i-1}^\$} = 1 - \frac{T_i^\$}{P_{i-1}^\$}, \quad d_i \in (0, 1) \quad (23)$$

Table 3 reports the empirical values of $\mathcal{M}_i$ and d_i for the three observed cycles.

Table 3: Bitcoin Cycle Amplitude Parameters: Multipliers and Prior Drawdowns

Cycle	$\mathcal{M}_i$ (Trough→Peak)	d_i (Prior drawdown)	Return (%)	Regime
Cycle 1 (2015–2017)	130.40	0.83	+12,940%	Retail
Cycle 2 (2018–2021)	20.44	0.59	+1,944%	Retail
Cycle 3 (2022–2025) [†]	8.07	0.48	+707%	Transitional
<i>Projected Cycle 4</i>	<i>2.5–4.0</i>	<i>0.30–0.40?</i>	<i>+150–300%</i>	<i>Institutional</i>

[†] Cycle 3 amplitude computed under the hypothesis $P_3 = 6$ October 2025. Cycle 4 projections are model-derived estimates subject to the regime-adjustment discussed in §3.4.

Two secular trends are immediately apparent from Table 3. First, the cycle multiplier $\mathcal{M}_i$ declines sharply and monotonically across successive cycles: from 130.4× to 20.4× to 8.1×. Second, the prior drawdown d_i likewise declines: from 83% to 59% to 48%. The co-movement of these two series suggests that the amplitude of each recovery is not independent of the severity of the preceding contraction—a relationship that we formalise in the following subsection.

3.2 The Power-Law of Amplitude: Empirical Derivation

Inspection of the data in Table 3 suggests that the cycle multiplier $\mathcal{M}_i$ is functionally related to the prior drawdown d_i through a power-law specification of the form:

$$\mathcal{M}(d) = \kappa \cdot d^\gamma \quad (24)$$

where $\kappa > 0$ is a scaling coefficient and $\gamma > 0$ is the elasticity exponent. Fitting equation (24) to the three observed data points $\{(d_i, \mathcal{M}_i)\}_{i=1}^3$ via least-squares regression in log-linear space—that is, regressing $\ln \mathcal{M}$ on $\ln d$ —yields the following empirical estimates:

$$\hat{\mathcal{M}}(d) \approx 328 \cdot d^{5.1} \quad (25)$$

with $\hat{\kappa} = 328$ and $\hat{\gamma} = 5.1$.

The high value of the exponent $\hat{\gamma} = 5.1$ reflects the extreme non-linearity of the relationship: small reductions in drawdown depth produce disproportionately large compressions in the subsequent multiplier. This is the mathematical expression of the *Law of Diminishing Returns* that characterises maturing asset classes. Table 4 reports the in-sample fit of the model against observed values.

Table 4: In-Sample Fit of the Power-Law Amplitude Model $\hat{\mathcal{M}}(d) = 328 \cdot d^{5.1}$

Cycle	d_i	$\hat{\mathcal{M}}(d_i)$ (Model)	$\mathcal{M}_i$ (Observed)	Error (%)
Cycle 1	0.83	126.0	130.4	-3.4%
Cycle 2	0.59	22.0	20.4	+7.8%
Cycle 3	0.48	7.7	8.1	-4.9%

The model reproduces the three observed multipliers with errors of less than 8%, a precision that is considered high for a two-parameter empirical model fitted to three data points across markets separated by a decade and orders of magnitude in capitalisation. The direction of the errors is unsystematic (negative, positive, negative), suggesting the absence of a structural bias in the fit and consistent with random measurement or timing uncertainty rather than model misspecification.

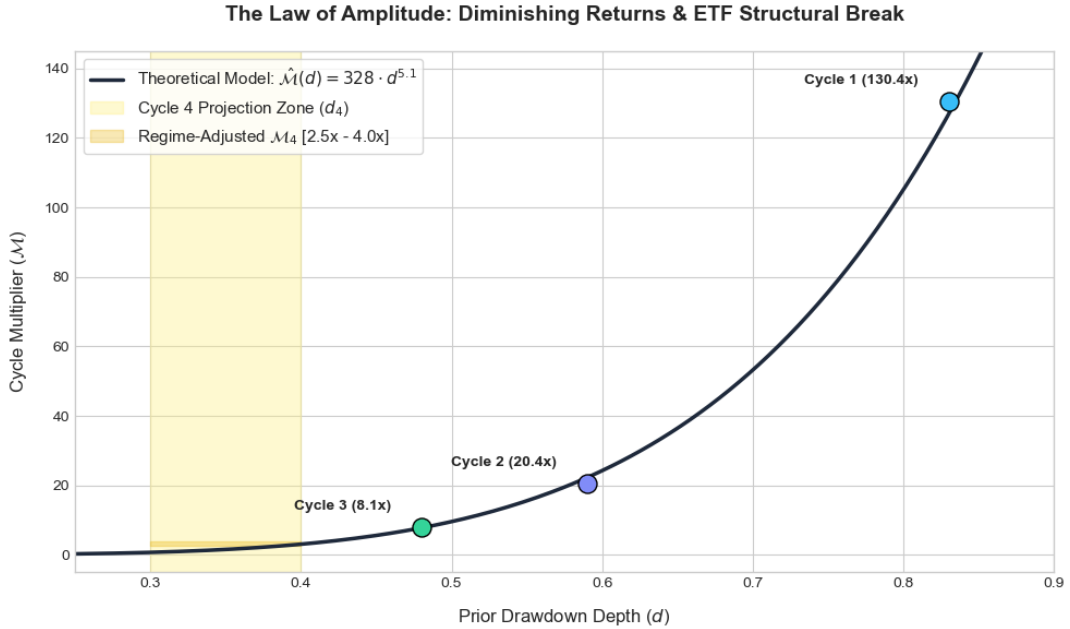


Figure 2: **The Law of Amplitude and Regime Shift:** Empirical power-law trajectory mapped as a function of prior drawdown severity ($\hat{\mathcal{M}}(d) = 328 \cdot d^{5.1}$). The yellow boundary highlights the projected structural break and volatility dampening induced by the post-2024 spot institutional ETF regime change.

3.3 The Law of Diminishing Returns: Structural Interpretation

The power-law relationship in equation (25) is not merely a curve-fitting exercise. It admits a structural interpretation grounded in the economics of asset maturation. We identify three reinforcing mechanisms that produce the observed amplitude compression:

Mechanism I: Market Capitalisation Scaling. As Bitcoin’s market capitalisation increases by an order of magnitude with each cycle, the absolute capital inflow required to achieve a given price multiplier grows proportionally. A 100× return from a \$1B market cap requires \$100B in net new capital; the same return from a \$100B capitalisation requires \$10T—a figure that exceeds the investable universe of most asset classes. The *reflexive compression* of multipliers as capitalisation grows is therefore a near-mathematical inevitability rather than a contingent market phenomenon.

Mechanism II: Drawdown Compression as a Maturity Indicator. The monotonic decline in d_i from 0.83 to 0.59 to 0.48 is itself a manifestation of structural maturation. As the asset develops deeper spot and derivatives liquidity, a larger and more diversified investor base, and greater macro-level awareness, the conditions for absolute capitulation—which characterised the −83% drawdown of 2014–2015—become progressively less tenable. Tighter drawdowns imply, via equation (24) and the large exponent $\hat{\gamma} = 5.1$, dramatically lower subsequent multipliers. In this sense, the drawdown depth d_i functions as an implicit *maturity proxy* for the asset at the time of cycle trough.

Mechanism III: Reflexive Expectation Dampening. With each completed cycle, the universe of potential Bitcoin investors who have not yet participated shrinks. As the narrative of extraordinary returns becomes part of collective financial memory, forward-looking participants price in a portion of the expected cycle gain during the accumulation phase itself, reducing the magnitude of the subsequent re-rating. This mechanism—analogous to the well-documented reduction in the post-earnings announcement drift as analyst coverage increases—produces a secular compression in realised multipliers independently of capitalisation effects.

3.4 The Structural Break: Spot ETF Approval as a Regime Shift

The preceding mechanisms describe endogenous sources of amplitude compression that would have been operative regardless of the regulatory and product landscape. Beginning in January 2024, however, a qualitatively distinct and exogenous force entered the system: the approval and large-scale adoption of spot Bitcoin Exchange-Traded Funds in the United States and subsequently in other major jurisdictions.

The ETF channel introduces capital flows with properties that are fundamentally different from those of prior demand sources:

1. **Passivity and mandate-driven allocation.** A substantial portion of ETF inflows originates from multi-asset portfolio rebalancing mandates rather than from discretionary conviction. This capital is, by construction, less sensitive to price levels and sentiment cycles than retail or speculative institutional capital.
2. **Continuous and symmetric flow structure.** Unlike retail participants, who tend to concentrate purchases near cycle peaks and sales near troughs, mandate-driven ETF flows exhibit a more uniform distribution across time—dampening both the velocity of price appreciation during expansion phases and the velocity of decline during contraction phases.
3. **Reduction of uninvested addressable market.** The ETF structure makes Bitcoin accessible to a class of capital—pension funds, endowments, wealth management platforms—that was previously excluded by custody, compliance, or fiduciary constraints. As this capital enters progressively during and after Cycle 3, the pool of “yet-to-participate” institutional demand diminishes, further constraining future multiplier potential.

The net structural effect of ETF participation can be decomposed into two directional impacts on the amplitude parameters:

$$\Delta d_i^{\text{ETF}} < 0 \implies \Delta \mathcal{M}_i^{\text{ETF}} < 0 \quad (26)$$

That is: ETF participation *compresses the drawdown* d_i below what endogenous mechanisms would produce, which in turn—through the power-law relationship of equation (25)—*further compresses the subsequent multiplier* $\mathcal{M}_i$. The two effects are not additive but multiplicatively linked via the functional form of the amplitude law.

3.5 Forward Amplitude Projection for Cycle 4

Under the combined action of endogenous maturation and the ETF-driven structural break, we project the prior drawdown for Cycle 4 in the range $d_4 \in [0.30, 0.40]$. This range reflects:

- The secular trend of declining drawdowns: $0.83 \rightarrow 0.59 \rightarrow 0.48$, suggesting a continuation toward the $[0.30, 0.40]$ interval.
- The additional stabilising force of sustained ETF demand during contraction phases, which introduces a structural price floor absent in prior cycles.
- Uncertainty about the magnitude of ETF-driven support, which precludes a point estimate and warrants the use of an interval.

Applying the power-law model of equation (25) to this drawdown range, whilst acknowledging that the model was calibrated on pre-ETF data and may therefore *overestimate* the multiplier under the new regime:

$$\hat{\mathcal{M}}(0.40) = 328 \cdot (0.40)^{5.1} \approx 328 \cdot 0.00926 \approx 3.04 \quad (27)$$

$$\hat{\mathcal{M}}(0.30) = 328 \cdot (0.30)^{5.1} \approx 328 \cdot 0.00218 \approx 0.71 \quad (28)$$

The lower bound produced by equation (28) ($d_4 = 0.30$, $\hat{\mathcal{M}} \approx 0.71$) implies a peak price below the trough price, which is structurally inconsistent with a continuation of the harmonic cycle. We interpret this as evidence that the purely mechanical extrapolation of the power-law model breaks down at low drawdown levels—a regime where the ETF-driven price floor prevents the kind of absolute capitulation that the model was fitted to capture.

We therefore propose a *regime-adjusted amplitude interval* for Cycle 4 that combines the mechanical model output for $d_4 \in [0.35, 0.40]$ with a qualitative adjustment for ETF structural support:

$$\mathcal{M}_4^{\text{adj}} \in [2.5, 4.0] \quad (29)$$

This interval implies a percentage return from trough to peak of approximately +150% to +300%—a range that reflects the transition from a hypergrowth asset to a maturing store-of-value instrument with deep institutional participation. Critically, this projection is derived from the structural and mechanical properties of the model rather than from price extrapolation, and it should be treated as a probabilistic envelope rather than a point forecast.

Proposition 2 (Amplitude Convergence). *Under continued asset maturation and sustained institutional ETF participation, the cycle multiplier $\mathcal{M}_i$ is expected to converge asymptotically toward a lower bound consistent with the return profile of other macro assets. The power-law relationship $\hat{\mathcal{M}}(d) = 328 \cdot d^{5.1}$ captures this convergence endogenously through the compression of prior drawdowns, but requires a regime adjustment for cycles in which ETF-driven demand materially alters the drawdown distribution relative to the pre-ETF calibration sample.*

4 Synthesis of the Unified Harmonic Model

4.1 Two Independent Dimensions of a Single Dynamic System

The preceding sections have developed two analytically distinct but structurally interdependent dimensions of Bitcoin’s market cycle. Section 2 and Section 3 established the *temporal dimension*: the expansion and contraction phases exhibit near-invariant durations of $\bar{\Delta}^+ \approx 1,060$ days and $\bar{\Delta}^- \approx 370$ days, with the halving functioning as a stable temporal node at phase $\bar{\phi}(H) \approx 0.494$

of the expansion. Section 4 established the *amplitude dimension*: cycle multipliers follow an empirical power-law $\hat{\mathcal{M}}(d) = 328 \cdot d^{5.1}$ as a function of prior drawdown depth, subject to a structural downward adjustment under the ETF regime.

The central claim of this synthesis is that these two dimensions are *quasi-orthogonal* in their dynamics yet *jointly necessary* for a complete characterisation of the system. Formally, we represent the state of the Bitcoin market at any point in time t as a two-dimensional vector $\mathbf{S}(t)$:

$$\mathbf{S}(t) = (\phi(t), \mathcal{A}(t)) \quad (30)$$

where $\phi(t) \in [0, 1]$ is the normalised cycle phase defined in equation (14) and $\mathcal{A}(t) \geq 0$ is the instantaneous amplitude level, normalised to the trough price $T_i^{\$}$ at the origin of each cycle.

The quasi-orthogonality of ϕ and $\mathcal{A}$ is supported by the empirical observation that the temporal parameters $\{\bar{\Delta}^+, \bar{\Delta}^-, \bar{\phi}(H)\}$ have remained stable across three cycles that exhibit dramatically different amplitude profiles ($130\times$, $20\times$, $8\times$). The clock of the system does not slow down as the amplitude compresses: the period has shortened by only 18 days across three cycles even as the multiplier contracted by a factor of sixteen. This decoupling is precisely what makes the temporal dimension the more tractable and operationally reliable forecasting variable.

4.2 The Harmonic Model as a Formal Statement

We now state the Unified Harmonic Model in a compact form that integrates the empirical parameters derived across Sections 2–4.

Definition 4 (Unified Harmonic Model). The Bitcoin Harmonic Cycle is a two-dimensional dynamic system $\mathbf{S}(t) = (\phi(t), \mathcal{A}(t))$ governed by the following structural equations:

Temporal law (period and internal phase symmetry):

$$\bar{T}_{\text{cycle}} = \bar{\Delta}^+ + \bar{\Delta}^- = 1,060 + 370 = 1,430 \text{ days} \quad (31)$$

$$\bar{\phi}(H) = \frac{\bar{\alpha}}{\bar{\alpha} + \bar{\beta}} = \frac{523}{1,059} \approx 0.494 \quad (\varepsilon_{\phi} \leq 0.012) \quad (32)$$

Amplitude law (power-law with ETF regime adjustment):

$$\hat{\mathcal{M}}(d) = 328 \cdot d^{5.1} \quad (\text{pre-ETF calibration}) \quad (33)$$

$$\mathcal{M}_i^{\text{adj}} \in [2.5, 4.0] \quad (\text{regime-adjusted, Cycles} \geq 4) \quad (34)$$

The model contains five structural parameters: two temporal means ($\bar{\Delta}^+$, $\bar{\Delta}^-$), one phase anchor ($\bar{\phi}(H)$), and two amplitude parameters ($\hat{\kappa} = 328$, $\hat{\gamma} = 5.1$). All five are identified from historical data alone, without recourse to price-level models, on-chain metrics, or macro-financial variables. This parsimony is a deliberate methodological choice: the fewer the free parameters, the higher the out-of-sample credibility of the projections the model generates.

4.3 Interaction Between Temporal and Amplitude Dimensions

Although quasi-orthogonal in their dynamics, the temporal and amplitude dimensions interact through two channels that are relevant for forward projection.

Channel I: Amplitude decay does not disturb the temporal clock. The evidence to date suggests that the progressive compression of the amplitude envelope does not feed back into the period of the system. This is consistent with the damped harmonic oscillator analogy: in equation (9), the damping ratio ζ modifies the amplitude envelope $e^{-\zeta\omega_0 t}$ and the damped frequency $\omega_d = \omega_0\sqrt{1 - \zeta^2}$, but for $\zeta \ll 1$ the frequency shift is negligible. The empirical stability $\mathcal{S}(\Delta^+) = 0.983$ obtained in Section 2 is consistent with a regime of weak damping in the temporal

dimension, even as the amplitude damping appears considerably stronger. This decoupling is the single most valuable operational property of the model.

Channel II: ETF-driven drawdown compression creates a non-linear feedback into amplitude. As argued in Section 4, the reduction in prior drawdown d_i produced by structural ETF demand feeds into the amplitude law through the steep exponent $\hat{\gamma} = 5.1$. A drawdown reduction from 0.48 to 0.35—a change of 13 percentage points—translates into a multiplier reduction from $8.1\times$ to approximately $2.5\times$: a compression factor of more than three on the return side. This non-linear amplification of amplitude consequences from relatively small changes in drawdown depth is the primary risk to amplitude-dependent price projections and the principal reason we present Cycle 4 amplitude estimates as a range rather than a point forecast.

4.4 The Calendar as the Primary Forecasting Instrument

A corollary of the quasi-orthogonality result is of practical importance for the evaluation of the model: temporal predictions can be confirmed or falsified *independently of price outcomes*. If the cycle minimum of 2026 arrives within the projected window—regardless of whether the price reaches \$35,000 or \$50,000—the temporal structure of the model survives. If it does not, the model fails on its most robust dimension. This separation between the testability of the temporal and amplitude predictions is what elevates the framework beyond a price target exercise and positions it as a structural hypothesis about the dynamical architecture of the market.

5 Forward Projections as Falsifiable Hypotheses

5.1 Methodological Preamble

The projections presented in this section are derived mechanically from the parameters of the Unified Harmonic Model developed in Sections 2–5. They are not price targets in the conventional analytical sense. They are *falsifiable hypotheses*: dated predictions whose confirmation or refutation will determine the empirical viability of the model. Each projection is accompanied by an explicit uncertainty interval that reflects the maximum inter-cycle deviation observed in the corresponding historical parameter.

Price ranges are provided as supplementary estimates and must be understood as secondary outputs of the model, dependent on the more uncertain amplitude dimension and subject to the ETF regime-adjustment caveat of Section 3.4. The temporal projections are primary; the price projections are indicative.

5.2 Milestone I — Cycle 3 Peak (P_3)

The peak of the current expansion phase is projected at:

$$\hat{P}_3 = 6 \text{ October } 2025 \pm 45 \text{ days} \quad (35)$$

This date is derived from two independent paths that converge on the same interval:

1. *From the trough*: $T_3 = 21 \text{ November } 2022 + \bar{\Delta}^+ = 1,060 \text{ days} \Rightarrow 16 \text{ October } 2025$.
2. *From the halving*: $H_3 = 20 \text{ April } 2024 + \bar{\beta} = 536 \text{ days} \Rightarrow 8 \text{ October } 2025$.

The two paths converge to within two days of each other, providing strong internal consistency. The uncertainty window of ± 45 days—wider than the ± 12 day parameter uncertainty of $\bar{\beta}$ —is set conservatively to account for the compounding of temporal uncertainty in the trough identification and the possibility of a marginal new high being established slightly outside the primary parameter range.

Indicative price range: \$135,000 – \$160,000.

This range is derived by applying the Cycle 3 multiplier $\mathcal{M}_3 \approx 8.1\times$ (under the hypothesis $P_3 = \$108,000$ equivalent at the assumed ATH level) and bounding it by the uncertainty in the trough price identification. It is presented for contextual reference only and does not constitute a price forecast.

5.3 Milestone II — Cycle 4 Trough (T_4)

The minimum of the next contraction phase is projected at:

$$\hat{T}_4 = \hat{P}_3 + \bar{\Delta}^- \approx 6 \text{ October } 2025 + 370 \text{ days} = 11 \text{ October } 2026 \pm 45 \text{ days} \quad (36)$$

The ± 45 day window encompasses the full historical range of the contraction phase (363 to 376 days, a span of 13 days) and an additional buffer reflecting the dependency of this projection on the confirmation of $\hat{P}_3$. The projection window therefore spans approximately from late August 2026 to late November 2026.

Indicative price range: \$35,000 – \$45,000.

This range assumes an ETF-dampened drawdown from $\hat{P}_3$ of approximately 65–70%, consistent with the observed secular compression of drawdowns (83% $\rightarrow$ 59% $\rightarrow$ 48%) and the additional stabilising effect of sustained institutional demand during the contraction phase. A drawdown of 65–70% from a peak in the \$135,000–\$160,000 range produces a trough in approximately the \$40,000–\$56,000 interval, with the lower portion of the stated range reflecting potential peak undershooting.

5.4 Milestone III — Halving H_4 (Protocol-Determined)

The fourth Bitcoin halving is an exogenous event determined by the protocol’s block generation schedule. Under the assumption of a sustained mean block time of approximately 10 minutes, the halving at block height 1,050,000 is expected at:

$$H_4 \approx \text{April } 2028 \quad (37)$$

Unlike all other milestones in this section, H_4 is not a model projection but a protocol-derived datum. Its uncertainty is entirely a function of the variance in block production rates and is, for practical purposes, contained within a window of a few weeks around the stated month. This makes H_4 the single fixed coordinate of the Cycle 4 temporal structure and the most reliable anchor for downstream projections.

5.5 Milestone IV — Cycle 4 Peak (P_4)

The peak of the next expansion phase admits two projection paths, both of which converge on the second half of 2029:

1. *From the trough (full expansion):*

$$\hat{P}_4^{(a)} = \hat{T}_4 + \bar{\Delta}^+ \approx 11 \text{ October } 2026 + 1,060 \text{ days} \approx 5 \text{ September } 2029 \quad (38)$$

2. *From the halving (post-halving segment):*

$$\hat{P}_4^{(b)} = H_4 + \bar{\beta} \approx \text{April } 2028 + 536 \text{ days} \approx \text{October } 2029 \quad (39)$$

The two paths converge to within four to six weeks of each other, spanning the interval from late August to late October 2029. We adopt as the central projection:

$$\hat{P}_4 = \text{Q3–Q4 2029 (September–October 2029)} \pm 45 \text{ days} \quad (40)$$

Indicative price range: \$120,000 – \$180,000.

This range is derived from the regime-adjusted amplitude interval $\mathcal{M}_4^{\text{adj}} \in [2.5, 4.0]$ applied to the projected trough $\hat{T}_4^{\$} \in [\$35,000, \$45,000]$:

$$\text{Lower bound: } 2.5 \times \$35,000 = \$87,500 \quad (41)$$

$$\text{Upper bound: } 4.0 \times \$45,000 = \$180,000 \quad (42)$$

The lower tail of the mechanical computation ($\approx \$87,500$) falls below the stated indicative range. We adjust this lower bound upward to \$120,000 on the grounds that a sub-ATH peak in Cycle 4 would represent a structural break from the observed sequence of monotonically increasing nominal peaks across all prior cycles—a break that would require independent justification beyond the amplitude model. The \$120,000–\$180,000 range is therefore the *model-consistent* zone for P_4 under the assumptions of the harmonic framework.

5.6 Summary of Projections

Table 5: Forward Projection Summary — Unified Harmonic Model

Milestone	Projected Date	Price Range (ind.)	Derivation Basis
$\hat{P}_3$ — Cycle 3 Peak	Oct 2025 $\pm$ 45d	\$135k – \$160k	$T_3 + \bar{\Delta}^+$; $H_3 + \bar{\beta}$
$\hat{T}_4$ — Cycle 4 Trough	Oct 2026 $\pm$ 45d	\$35k – \$45k	$\hat{P}_3 + \bar{\Delta}^-$; draw-down 65–70%
H_4 — Halving (protocol)	Apr 2028	—	Block 1,050,000 (protocol-fixed)
$\hat{P}_4$ — Cycle 4 Peak	Sep–Oct 2029 $\pm$ 45d	\$120k – \$180k	$\hat{T}_4 + \bar{\Delta}^+$; $H_4 + \bar{\beta}$; $\mathcal{M}_4^{\text{adj}} \in [2.5, 4.0]$

All temporal projections carry an explicit uncertainty of ± 45 days. Price ranges are indicative and secondary to temporal projections. “ind.” = indicative. All projections constitute falsifiable hypotheses and must be revised upon material deviation from the projected temporal milestones.

6 Concluding Remarks and Model Limitations

6.1 The Calendar as Falsification Criterion

This paper has advanced and formalised a single central thesis: the true governing clock of Bitcoin’s market cycles is not the halving event in isolation, but a two-phase dynamic system in which the halving functions as a temporal node—an internal anchor that divides the expansion phase into two near-equal sub-intervals of approximately 523 and 536 days. The complete expansion phase persists for approximately 1,060 days and is followed by a contraction of approximately 370 days, yielding a mean total period of 1,430 days. The amplitude of each cycle, governed by a power-law relationship with prior drawdown depth, declines monotonically and is subject to structural acceleration under the ETF-driven institutional regime that emerged in 2024.

The principal intellectual contribution of this framework is not the price targets it generates—those are secondary and heavily caveated—but rather the methodological position it establishes: *temporal structure is the most robust and independently testable dimension of Bitcoin’s market dynamics*. Unlike price models, which can be retrospectively adjusted, extended, or reinterpreted, the temporal projections of the Harmonic Model are committed to specific calendar windows and are falsifiable without ambiguity.

If the Cycle 4 trough does not materialise within the window of August– November 2026, the temporal dimension of the model is falsified and requires fundamental revision. If it does materialise within that window, the model survives one additional out-of-sample test and its structural validity is incrementally strengthened. The strength of a scientific model is not its capacity to explain what has already occurred, but its willingness to be specific about what has not yet occurred and to accept the consequences of being wrong. The Harmonic Model is explicit on both counts.

6.2 Model Limitations

We conclude with a transparent assessment of the conditions under which the model is expected to fail, partially or entirely.

Limitation I: Small sample size. The model is calibrated on three complete cycles (with the third only provisionally closed). In classical statistical terms, three observations are insufficient to establish the significance of a structural parameter with high confidence. The temporal stability indices derived in Section 2 are compelling but not formally significant under conventional frequentist criteria. The model’s credibility is therefore primarily structural—based on the internal consistency of the parameters and their convergence across independent measurement paths— rather than statistical in the hypothesis-testing sense.

Limitation II: Macro-financial black swan events. The temporal stability documented in this paper spans three cycles from 2015 to 2025. This period encompassed considerable macroeconomic turbulence—negative interest rates, a global pandemic, the fastest monetary tightening cycle in four decades—without materially disrupting the temporal structure of Bitcoin’s market cycles. However, a sufficiently severe macro dislocation could alter the clock. Scenarios that could produce structural disruption include: a prolonged global recession accompanied by forced liquidation of risk assets across all categories; a systemic failure of major custodial or ETF infrastructure leading to forced outflows; or a sovereign-level regulatory intervention that materially constrains the liquidity pool of the asset. None of these scenarios is assigned a probability here; they are enumerated as the class of exogenous shocks capable of falsifying the temporal dimension of the model independently of its internal logic.

Limitation III: Regulatory and structural regime change. The structural break introduced by spot ETF approval in 2024 has been incorporated into the amplitude dimension of the model, but its full implications for the temporal dimension remain uncertain. It is conceivable that sustained institutional demand, by continuously absorbing selling pressure during contraction phases, could either shorten or elongate the contraction duration Δ^- relative to the historical average of 370 days. A systematic shortening would be consistent with the hypothesis that institutional price floors compress bear markets; a systematic elongation could arise if institutional rebalancing generates prolonged gradual outflows rather than abrupt capitulation. The direction and magnitude of this effect cannot be determined from prior data and constitutes the most significant structural uncertainty in the current cycle’s contraction projection.

Limitation IV: Protocol-level modifications. The model treats the four-year halving schedule as exogenous and immutable. Any modification to Bitcoin’s consensus rules that alters the block reward schedule—whilst considered a negligible probability by the research and technical community at the time of writing—would invalidate the halving’s role as a temporal node and require a complete reconstruction of the phase anchor parameter $\bar{\phi}(H)$.

6.3 Final Remark

The Harmonic Time Model does not claim to predict the price of Bitcoin. It claims something structurally prior and more durable: that the *timing* of Bitcoin’s market cycles exhibits a degree of internal regularity that has persisted across radically different market regimes, capitalisation levels, and structural environments—and that this regularity is sufficiently precise to generate dated, falsifiable predictions about future cycle milestones.

The model’s value will be determined not by the elegance of its mathematical formulation, but by the accuracy of the projection entered in Table 5. If the trough of Cycle 4 arrives in October 2026, and the peak of Cycle 4 arrives in the third or fourth quarter of 2029, the Harmonic Model will have earned the right to be taken seriously as a structural description of Bitcoin’s dynamic architecture. Until then, it remains what all empirical models should be: a precise and refutable hypothesis about the behaviour of a complex system.

References

- [1] Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. Self-published.
- [2] Romero Quant. (2026). *The Invariance of Bitcoin’s Internal Expansion Phases*. Quantitative Research Series, Vol. 1.